HW 11

Minh Truong

141225

Perform the following activities in Python:

- Get the dataset - pick any one of the datasets from the NLP Datasets folder or from any other online resource that interests you.
- Load the .csv file into a data frame in Python.
- You can use any package of your choice in Python - NLTK, VADER etc.,
- Import necessary libraries.
- Preprocess data if needed.
- Perform Sentiment Analysis.
- Classify the information into Positive, Negative, or Neutral sentiments.
- Classify the text categorically into emotions - anger, disgust, fear, joy, sadness, and surprise etc.,
- Generate bar graphs for sentiment and emotion.
- Generate a word cloud.
- Provide conclusions based on the analysis.

Import necessary libraries.

In [31]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from wordcloud import WordCloud
from nrclex import NRCLex
import nltk
import re
import subprocess
subprocess.run(['python', '-m', 'textblob.download_corpora'])

# Download required NLTK resources
nltk.download('vader_lexicon')
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     /Users/mtruong/nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     /Users/mtruong/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt to /Users/mtruong/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     /Users/mtruong/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

Out[31]: True

Get the Dataset

Load the .csv file into a data frame in Python.

In [16]:
```
# Load dataset
df = pd.read_csv('/Users/mtruong/python-projects/BZAN 6355/Original recipes.
df.head()
```

Out[16]:

	name	id	minutes	contributor_id	submitted	tags	nutrition	n_st
0	arriba baked winter squash mexican style	137739	55	47892	9/16/2005	['60-minutes-or-less', 'time-to-make', 'course...	[51.5, 0.0, 13.0, 0.0, 2.0, 0.0, 4.0]	
1	a bit different breakfast pizza	31490	30	26278	6/17/2002	['30-minutes-or-less', 'time-to-make', 'course...	[173.4, 18.0, 0.0, 17.0, 22.0, 35.0, 1.0]	
2	all in the kitchen chili	112140	130	196586	2/25/2005	['time-to-make', 'course', 'preparation', 'mai...	[269.8, 22.0, 32.0, 48.0, 39.0, 27.0, 5.0]	
3	alouette potatoes	59389	45	68585	4/14/2003	['60-minutes-or-less', 'time-to-make', 'course...	[368.1, 17.0, 10.0, 2.0, 14.0, 8.0, 20.0]	
4	amish tomato ketchup for canning	44061	190	41706	10/25/2002	['weeknight', 'time-to-make', 'course', 'main-...	[352.9, 1.0, 337.0, 23.0, 3.0, 0.0, 28.0]	

Describe the Data

In [17]:

```
# Summary statistics
df.describe()
```

Out[17]:

	id	minutes	contributor_id	n_steps	n_ingredients
count	231637.000000	2.316370e+05	2.316370e+05	231637.000000	231637.000000
mean	222014.708984	9.398546e+03	5.534885e+06	9.765499	9.051153
std	141206.635626	4.461963e+06	9.979141e+07	5.995128	3.734796
min	38.000000	0.000000e+00	2.700000e+01	0.000000	1.000000
25%	99944.000000	2.000000e+01	5.690500e+04	6.000000	6.000000
50%	207249.000000	4.000000e+01	1.736140e+05	9.000000	9.000000
75%	333816.000000	6.500000e+01	3.982750e+05	12.000000	11.000000
max	537716.000000	2.147484e+09	2.002290e+09	145.000000	43.000000

In [18]:

```
# Check for missing values
df.isnull().sum()
```

Out[18]:

```
name                 1
id                   0
minutes              0
contributor_id       0
submitted            0
tags                 0
nutrition            0
n_steps              0
steps                0
description       4979
ingredients          0
n_ingredients        0
dtype: int64
```

Preprocess data if needed.

- Drop rows with missing descriptions
- Sample 5,000 rows for performance
- Clean text

In [19]:

```
# Drop missing descriptions and sample 5,000 rows for performance
df = df[['name', 'description']].dropna().sample(5000, random_state=42).rese

# Clean text by lowercase text and removing punctuation/numbers
def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    return text.strip()

df['clean_description'] = df['description'].apply(clean_text)
df.head()
```

Out[19]:

	name	description	clean_description
0	auntie fanny s pea and mint mash	this is still only peas, tastes of peas, and d...	this is still only peas tastes of peas and doe...
1	honey roasted plums with thyme	excellent, elegant desert, especially when ser...	excellent elegant desert especially when serve...
2	egg muffin sandwich	just like mcdonalds, but you can control the i...	just like mcdonalds but you can control the in...
3	splendid chicken and corn	i tried a recipe very similar to this and love...	i tried a recipe very similar to this and love...
4	applesauce raisin muffins	this muffin recipe is a hit with all my family...	this muffin recipe is a hit with all my family...

Perform Sentiment Analysis.

Classify the information into Positive, Negative, or Neutral sentiments.

Decided to use VADER because it is designed for short, informal text and scores each description as positive, negative, or neutral based on compound score.

In [32]:

```
# Initialize VADER sentiment analyzer
sia = SentimentIntensityAnalyzer()

# Compute compound sentiment score for each description
# compound ranges from -1 (negative) to +1 (positive)
df['compound'] = df['clean_description'].apply(lambda x: sia.polarity_scores

# Apply standard VADER thresholds to label sentiment
def classify_sentiment(score):
    if score >= 0.05:
        return 'Positive'
    elif score <= -0.05:
        return 'Negative'
    else:
        return 'Neutral'

df['sentiment'] = df['compound'].apply(classify_sentiment)
df[['description', 'sentiment', 'compound']].head(10)
```

Out[32]:

	description	sentiment	compound
0	this is still only peas, tastes of peas, and d...	Positive	0.7962
1	excellent, elegant desert, especially when ser...	Positive	0.7783
2	just like mcdonalds, but you can control the i...	Positive	0.8126
3	i tried a recipe very similar to this and love...	Positive	0.9246
4	this muffin recipe is a hit with all my family...	Positive	0.8360
5	for this recipe the macaroni is cooked right i...	Positive	0.1406
6	quick and tasty way to serve chicken breasts. ...	Positive	0.2732
7	this can also be made using chicken pieces or ...	Positive	0.9563
8	from publix (supermarket)apron's, simple meal ...	Positive	0.5859
9	always a crowd pleaser, and kid pleaser too!\r...	Positive	0.9300

Classify Emotions

Classify the text categorically into emotions - anger, disgust, fear, joy, sadness, and surprise etc.,

In [47]:

```
# Tokenize text and sum NRC emotion scores across all words
# Removes 'positive'/'negative' because these are sentiment labels, not emot
def get_emotion(text):
    try:
        words = word_tokenize(str(text).lower())
        scores = {}
        for word in words:
            emotion_obj = NRCLex(word)          # Initialize NRCLex with sir
            emotion_obj.load_raw_text(word)     # Explicitly load word as ra
            for k, v in emotion_obj.affect_frequencies.items():
                if v > 0:
                    scores[k] = scores.get(k, 0) + v  # Accumulate emotion s
        scores.pop('positive', None)
        scores.pop('negative', None)
        if not scores or max(scores.values()) < 0.5:  # Ignore weak signals
            return 'none'
        return max(scores, key=scores.get)             # Return dominant emo
    except Exception:
        return 'none'

df['emotion'] = df['clean_description'].apply(get_emotion)
df[['description', 'sentiment', 'emotion']].head(10)
```

Out[47]:

	description	sentiment	emotion
0	this is still only peas, tastes of peas, and d...	Positive	none
1	excellent, elegant desert, especially when ser...	Positive	none
2	just like mcdonalds, but you can control the i...	Positive	none
3	i tried a recipe very similar to this and love...	Positive	none
4	this muffin recipe is a hit with all my family...	Positive	none
5	for this recipe the macaroni is cooked right i...	Positive	none
6	quick and tasty way to serve chicken breasts. ...	Positive	none
7	this can also be made using chicken pieces or ...	Positive	none
8	from publix (supermarket)apron's, simple meal ...	Positive	none
9	always a crowd pleaser, and kid pleaser too!\r...	Positive	none

In [68]:

```
def get_emotion(text):
    try:
        word_list = word_tokenize(str(text).lower())
        scores = {}
        lexicon = NRCLex('a').__lexicon__
        for word in word_list:
            if word in lexicon:
                # Lexicon values are lists of emotion strings, not dicts
                for emotion in lexicon[word]:
                    if emotion not in ('positive', 'negative'):
                        scores[emotion] = scores.get(emotion, 0) + 1
        if not scores:
            return 'none'
        return max(scores, key=scores.get)
    except Exception:
        # Suppress error output to prevent thousands of repeated print lines
        return 'none'

df['emotion'] = df['clean_description'].apply(get_emotion)
print(df['emotion'].value_counts())
```

```
emotion
joy             1533
anticipation    1226
none             952
trust            725
fear             166
anger            162
disgust          103
sadness           89
surprise          44
Name: count, dtype: int64
```

In [69]:

```
lexicon = NRCLex('a').__lexicon__
# Check what a single entry looks like
```

```
sample_key = list(lexicon.keys())[0]
print(sample_key, lexicon[sample_key])
```

```
abacus ['trust']
```

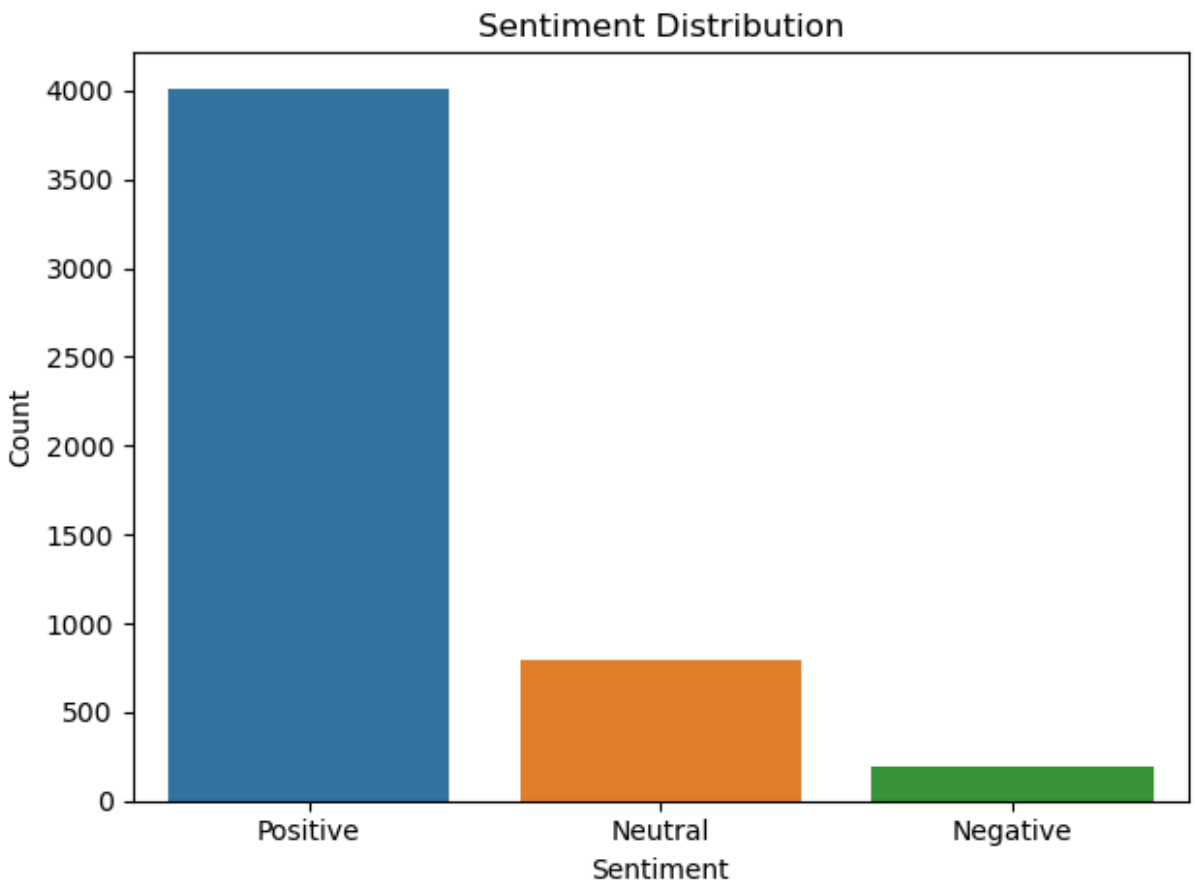
In [70]:
```
def get_emotion(text):
    try:
        word_list = word_tokenize(str(text).lower())
        scores = {}
        lexicon = NRCLex('a').__lexicon__
        for word in word_list:
            if word in lexicon:
                for emotion in lexicon[word]:                        # List
                    if emotion not in ('positive', 'negative'):
                        scores[emotion] = scores.get(emotion, 0) + 1 # Count
        if not scores:
            return 'none'
        return max(scores, key=scores.get)
    except Exception as e:
        print(e)
        return 'none'

df['emotion'] = df['clean_description'].apply(get_emotion)
print(df['emotion'].value_counts())
```

```
emotion
joy             1533
anticipation    1226
none             952
trust            725
fear             166
anger            162
disgust          103
sadness           89
surprise          44
Name: count, dtype: int64
```

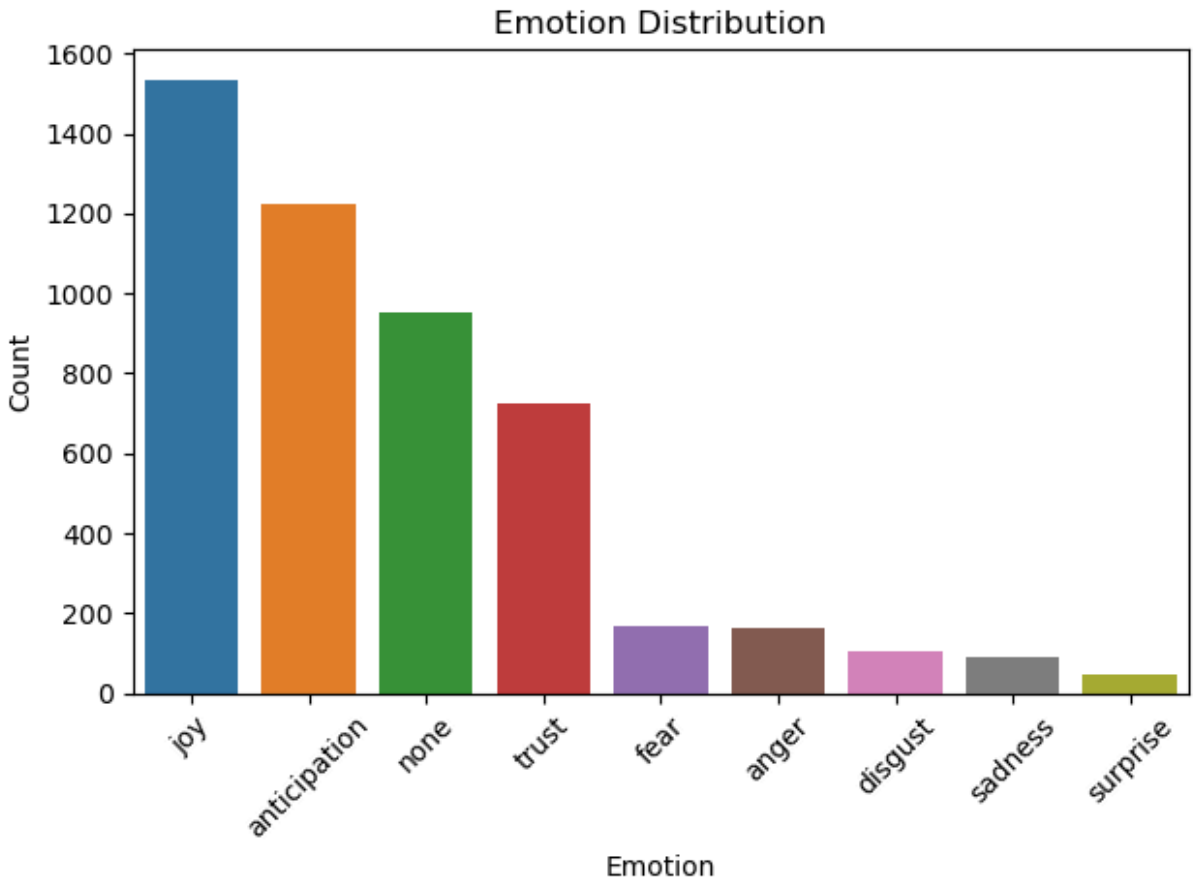
Generate bar graphs for sentiment and emotion.

In [71]:
```
# Sentiment Distribution Bar Graph
# Count occurrences of each sentiment label
sentiment_counts = df['sentiment'].value_counts().reset_index()
sentiment_counts.columns = ['sentiment', 'count']

sns.barplot(
    data=sentiment_counts,
    x='sentiment',
    y='count',
    hue='sentiment', # Colors bars by sentiment category
    palette='tab10',
    legend=False # Suppress redundant legend
)
plt.title('Sentiment Distribution')
plt.xlabel('Sentiment')
plt.ylabel('Count')
```

```
plt.tight_layout()
plt.show()
```



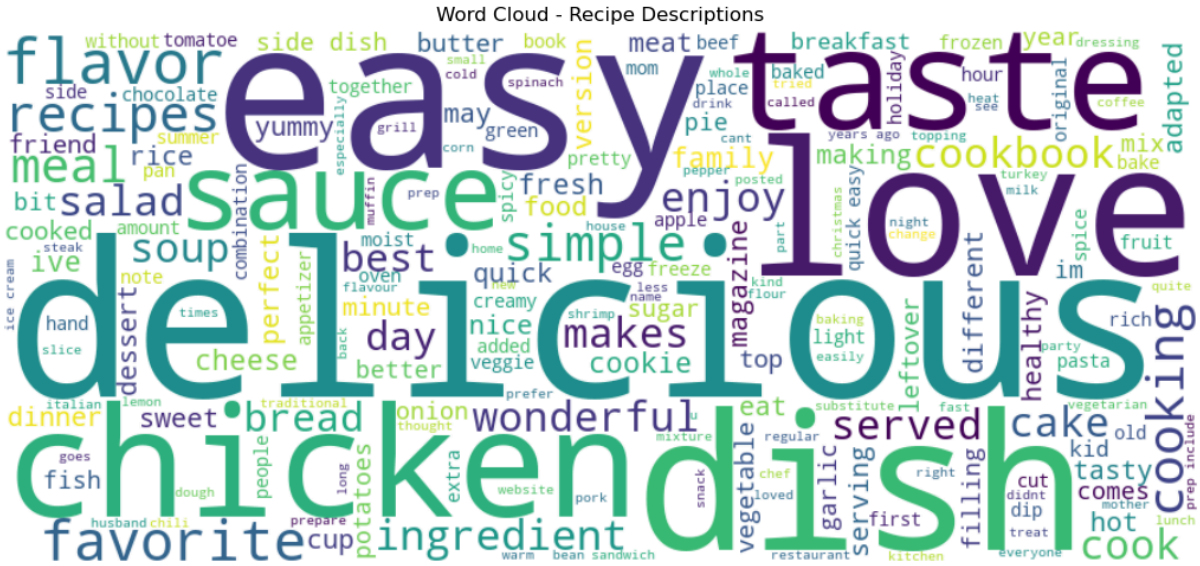
In [72]:
```
# Emotion Distribution Bar Graph
# Count occurrences of each emotion label
emotion_counts = df['emotion'].value_counts().reset_index()
emotion_counts.columns = ['emotion', 'count']

sns.barplot(
    data=emotion_counts,
    x='emotion',
    y='count',
    hue='emotion', # Colors bars by emotion category
    palette='tab10',
    legend=False # Suppress redundant legend
)
plt.title('Emotion Distribution')
plt.xlabel('Emotion')
plt.ylabel('Count')
plt.xticks(rotation=45) # Rotate labels to prevent overlap
plt.tight_layout()
plt.show()
```



Generate a word cloud.

In [73]:
```
# Word cloud (most frequent words in all descriptions)
stop_words = set(stopwords.words('english'))

# Add domain-specific filler words that add no analytical value
custom_stopwords = stop_words | {'recipe', 'make', 'use', 'add', 'get',
                                  'want', 'one', 'also', 'well', 'even',
                                  'really', 'little', 'dont', 'used', 'made'
                                  'using', 'time', 'way', 'much', 'many',
                                  'would', 'could', 'like', 'great', 'good',
                                  'need', 'go', 'try', 'work', 'look',
                                  'serve', 'found', 'came', 'think', 'got',
                                  'put', 'come', 'find', 'take', 'say',
                                  'know', 'give', 'call', 'keep', 'still',
                                  'two', 'every', 'another', 'something',
                                  'sure', 'lot', 'never', 'always', 'alway',
                                  'thing', 'instead', 'usually', 'enough'}

all_text = ' '.join(df['clean_description'])

# Remove stopwords and filler words to surface meaningful terms
filtered_text = ' '.join([w for w in all_text.split() if w not in custom_stc

# Generate and display word cloud
# The larger words are the ones that appear more frequently
wordcloud = WordCloud(width=900, height=400, background_color='white').gener
```

```
plt.figure(figsize=(12, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Word Cloud - Recipe Descriptions')
plt.tight_layout()
plt.show()
```


Word Cloud - Recipe Descriptions
delicious
chicken
easy
love
taste
flavor
sauce
recipes
meal
salad
soup
dish
favorite
ingredient
wonderful
simple
cookbook
cooking
cook
served
cake
bread
makes
enjoy
family
fresh
food
best
day
cheese
butter
side dish
meat
breakfast
baked
pie
yummy
may
green
together
chocolate
friend
rice
cooked
bit
ive
dessert
perfect
appetizer
combination
version
quick
nice
minute
creamy
sugar
cookie
top
eat
vegetable
garlic
serving
filling
first
leftover
different
healthy
pasta
rich
fruit
spice
mix
adapted
original
year
frozen
holiday
hour
place
making
fish
potatoes
cup
sweet
dinner
onion
tasty
comes
hot
kid
old
cut
dip
pork
chicken
beef
tomatoe
without
spinach
cold
small
mom
whole
called
tried
drink
heat
see
coffee
years ago
topping
pepper
posted
turkey
milk
bake
night
change
christmas
quick easy
part
house
apple
egg
freeze
less
name
kind
flour
shrimp
baking
easily
substitute
fast
vegetarian
regular
mixture
thought
prefer
traditional
italian
lemon
goes
dough
husband
chili
prepare
website
long
extra
people
muffin
especially
grill
corn
prep
spicy
pretty
summer
pan
note
amount
steak
ice cream
hand
slice
times
back
oven
moist
flavour
home
new
warm
bean sandwich
restaurant
kitchen
everyone
chef
loved
snack
right
didnt
treat
mother
lunch
prep include
party
quite
dressing
coffee
topping